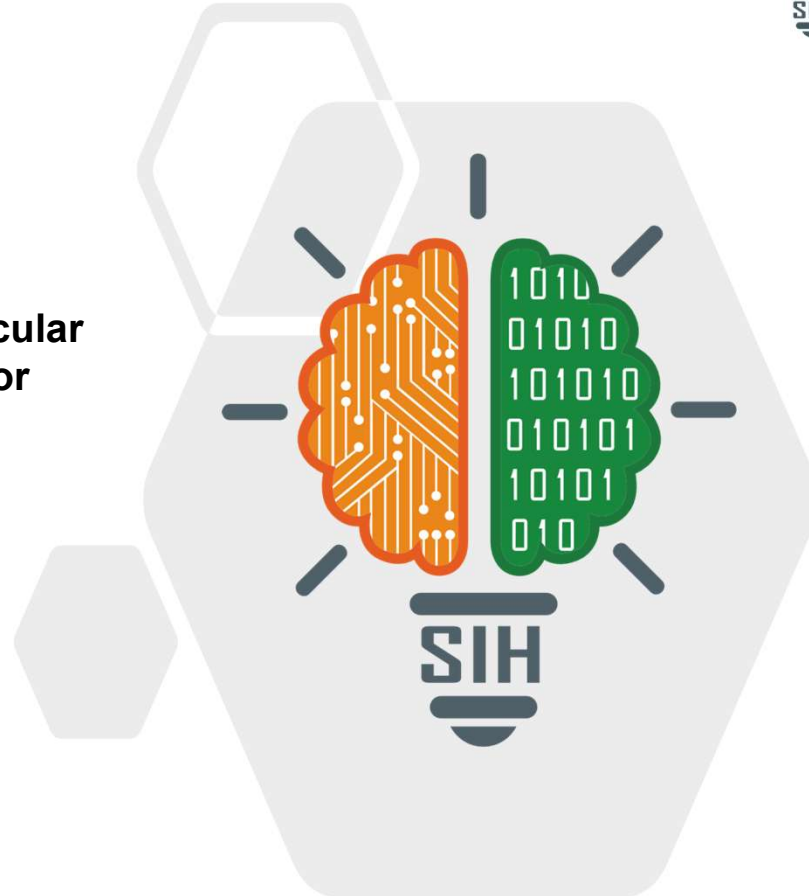


SMART INDIA HACKATHON 2026



- **Problem Statement ID – SIH26042**
- **Problem Statement Title- AI-Powered Vernacular Pedagogy and Real-Time Translation Tool for Mother Tongue-Based Primary Education**
- **Theme- Smart Education**
- **PS Category- Software**
- **Team ID-**
- **Team Name- PROBLEM SOLVER**



Proposed Solution

1 TEACHER INPUT

Select Grade + Subject + Mother
Tongue
Enter the day's teaching requirement.

2 AI GENERATION

Google Gemini-AI powered contextual
generation
with structured teacher prompts.

3 READY TO USE

Mother-tongue lesson, worksheet and
3-question assessment for the class.

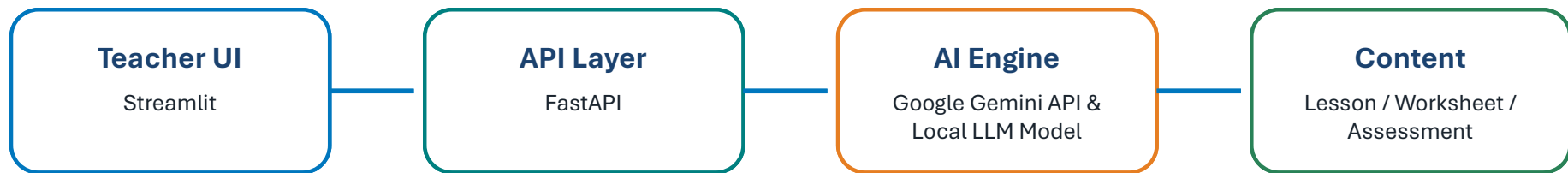
How it addresses the problem

- Reduces Hindi/English ↔ mother-tongue comprehension gap
- Uses familiar local objects and classroom contexts
- Reduces teacher preparation time
- Supports daily FLN practice and assessment

Innovation & uniqueness

- **Contextual AI:** Uses local objects (mangoes, trees, goats) for tribal FLN, not abstract examples.
- **Teacher-First:** Instantly generates printable mother-tongue worksheets, assessments, and lessons.
- **Offline-Ready Architecture:** Streamlit + FastAPI + Gemini, built for future offline mobile deployment.

Technical Approach



Implementation methodology

- Structured prompt → AI inference → formatted output
- HTTP POST from Streamlit to FastAPI
- Backend validates request and returns generated content

Technology stack

- **Frontend:** Streamlit, HTML, CSS
- **Backend:** Python, FastAPI, Uvicorn, Pydantic, Requests
- **AI Engine:** Google Gemini (Flash), Prompt Engineering
- **Architecture:** REST API, CORS Middleware, Localhost Hosting

AI/ML note: the prototype uses an external LLM for inference; it does not require training a custom ML model.



FEASIBILITY AND VIABILITY



Feasibility & Viability

FEASIBLE

Cloud AI avoids expensive local AI hardware.
Python stack is deployable on standard systems.

CHALLENGES

Rural connectivity
Limited verified tribal-language data
LLM output quality / consistency

MITIGATION

Offline mobile/tablet roadmap
Verified corpus + human validation
Prompt constraints + evaluation

Viability path



Key viability principle: start cloud-first, validate language quality, then progressively add offline capability.



IMPACT AND BENEFITS



Impact & Benefits

STUDENTS

Better comprehension
Mother-tongue reinforcement
Stronger FLN practice

TEACHERS

Faster lesson preparation
Ready assessments
Printable classroom material

COMMUNITY

Supports tribal languages
Culturally relevant learning
Greater inclusion

SYSTEM

Scalable digital workflow
Potential district analytics
Lower paper wastage

Alignment

NEP 2020

NIPUN Bharat / FLN

Tribal-language
inclusion

Teacher
productivity

Expected outcome → more understandable instruction, faster teacher preparation, and stronger mother-tongue participation in FLN learning.

Research & References

01

National Education Policy (NEP) 2020

Ministry of Education, Government of India

02

NIPUN Bharat Mission Guidelines

Ministry of Education, Government of India

03

ASER Reports (2023/2024)

Rural and foundational learning context

04

DeepSeek API / LLM Documentation

AI inference and integration reference

05

Streamlit • FastAPI • Python

Official technical documentation

RESEARCH TAKEAWAY

The solution combines mother-tongue support, teacher assistance and AI generation in one classroom workflow. Language quality must be validated with native speakers/corpus before production-scale deployment.

BhashaSetu

Bridging language and learning.